The first excitation energy is

$$E_I = E_2 - E_1 = -13.6\,eV - (-54.4\,eV)$$
$$= 40.8\,eV$$

Hence, the first excitation potential is

$$V_I = \frac{1}{e}E_I = \frac{(40.8\,eV)}{e}$$
$$= 40.8\,\text{volt}$$

The first ionization energy is

$$E_{\text{ionization}} = E_\infty - E_1 = 0 - (-54.4\,eV)$$
$$= 54.4\,eV$$

Hence, the first ionization potential is

$$V_{\text{ionization}} = \frac{1}{e}E_{\text{ionization}} = \frac{(54.4\,eV)}{e}$$
$$= 54.4\,\text{volt}$$

(c) Consider two photons to be A and B.
Given that photon A with energy 42 eV and photon B with energy 51 eV
From Bohr assumption, difference in energy levels is equal to the energy photon absored, then atom will absorb energy, otherwise, not.

$$E_2 - E_1 = -13.6\,eV - (-54.4\,eV)$$
$$= 40.8\,eV \approx 41\,eV$$

Similarly,

$$E_3 - E_1 = -6.04\,eV - (-54.4\,eV)$$
$$= 48.36\,eV$$

$$E_4 - E_1 = -3.4\,eV - (-54.4\,eV)$$
$$= 51\,eV$$

$$E_3 - E_2 = -6.04\,eV - (-13.6\,eV)$$
$$= 7.56\,eV$$

and so on.

But note that $E_2 - E_1 \neq 42$ eV, $E_3 - E_1 \neq 42$ eV, $E_4 - E_1 \neq 42$ eV and $E_3 - E_2 \neq 42$ eV.

For all possibilities, no difference in energy is an integer multiple of photon energy. Hence, photon A is not absorbed by this atom. But for Photon B, $E_4 - E_1 = 51$ eV, which means, Photon B can be absorbed by this atom.

(d) The radius of Bohr orbit is $r_n = \frac{a_0 \times n^2}{z}$

For $n = 1$, $z = 2$

$$r_1 = \frac{a_0}{2}$$
$$= \frac{0.529}{2}$$
$$= 0.265\ \text{Å}$$

(e) Since, total energy is equal to negative of kinetic energy in Bohr atom model, we get

$$KE_n = -E_n = -\left(-\frac{54.4}{n^2}eV\right)$$
$$= \frac{54.4}{n^2}eV$$

Since, Potential energy is negative of twice the kinetic energy,

$$U_n = -2KE_n = -2\left(\frac{54.4}{n^2}eV\right)$$
$$= -\frac{108.8}{n^2}eV$$

For a ground state, put n =1

Kinetic energy is KE_1 = 54.4 eV and Potential energy is U_1 = –108.8 eV

9.3.4 Atomic spectra

Materials in the solid, liquid and gaseous states emit electromagnetic radiations when they are heated up and these emitted radiations usually exhibit continuous spectrum. For example, when white light is examined through a spectrometer, electromagnetic radiations of all wavelengths are observed which is a continuous spectrum.

In early twentieth century, many scientists spent considerable time in understanding the characteristic radiations emitted by the atoms of individual elements exposed to a flame or

UNIT-9(XII-Physics_Vol-2)_New.indd 160 21-07-2020 15:48:52